

— TECHNICAL CASE STUDY

Cloud Storage using **Amazon S3**

Designing a secure and scalable storage layer for a College Management System — from local hard drives to cloud-native object storage.



PRESENTED BY
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Why Does a College Need Cloud Storage?

A college management system never stops generating files — but saving them on physical local servers creates massive risks.

DAILY COLLEGE DATA FLOW

Thousands of Files / Week



Student & Academic Records

MARKSHEETS / HIGH VOLUME

Marksheets, degree certificates, assignment submissions, attendance PDFs



Identity & Confidential Data

SENSITIVE / PRIVATE

Aadhaar cards, student KYC scans, faculty payroll, official fee receipts



Heavy Digital Media

GIGABYTES / FILE

HD lecture video recordings, convocation photos, annual campus fest media

The diagram illustrates a single point of failure in a cloud architecture. It features a dark blue background with a light blue grid. At the top, a red banner with a white warning triangle icon contains the text "BEFORE CLOUD: THE PHYSICAL BOTTLENECK". To the right of this banner is a red pill-shaped button with the text "HIGH RISK". Below the banner, three light blue rounded rectangular boxes are arranged vertically, connected by thin vertical lines. The first box is labeled "Student / Faculty" with a graduation cap icon and contains the text "Requests or uploads file from browser". The second box is labeled "College Web Server" with a server rack icon and contains the text "Proxies all file traffic through single host CPU". The third box is labeled "Local Mechanical HDD" with a hard drive icon and contains the text "Raw internal disk — zero automated redundancy". To the right of the third box, the text "SINGLE POINT OF FAILURE" is written in large, bold, red capital letters.

BEFORE CLOUD: THE PHYSICAL BOTTLENECK HIGH RISK

Student / Faculty
Requests or uploads file from browser

College Web Server
Proxies all file traffic through single host CPU

Local Mechanical HDD
Raw internal disk — zero automated redundancy

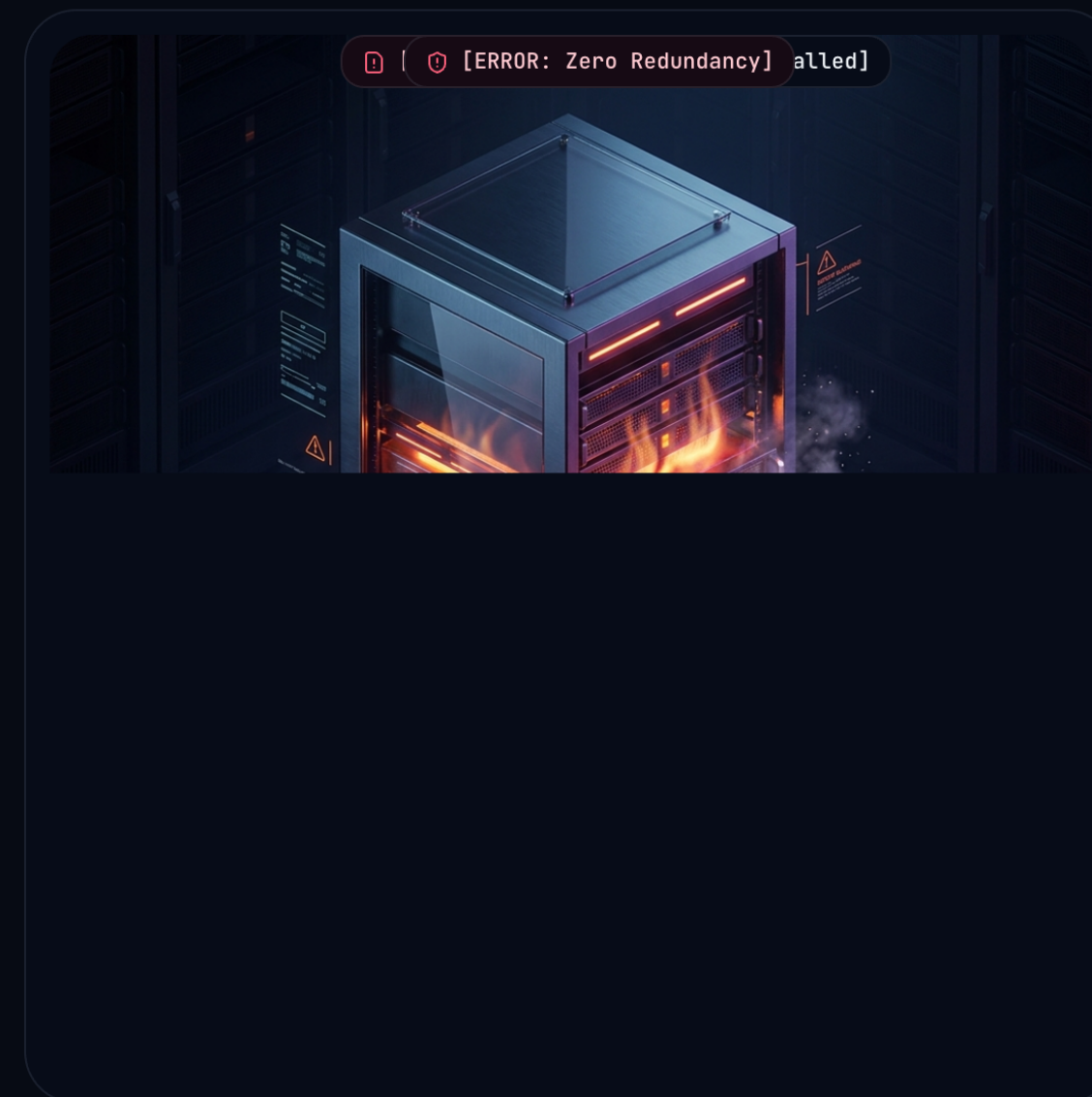
SINGLE POINT OF FAILURE

 TECHNICAL DEEP DIVE

 Basic

 Technical

A college produces thousands of documents, photos and videos every day. Storing them on an ordinary office computer or server is risky and quickly runs out of space.



Object Storage Built for the Internet

Amazon S3 (Simple Storage Service) stores and retrieves any amount of data from anywhere over the internet.

Instead of files living on the college's own hard drives, they become **objects** inside an **S3 bucket** — while the application keeps working exactly as before.

TRADITIONAL STORAGE

Server + HDD + Backup +
Maintenance

Buy hardware upfront · capacity fixed ·
you manage failures

AMAZON S3

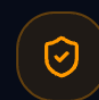
App → S3 Bucket

Unlimited capacity · AWS-managed
durability · pay per use



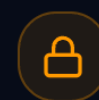
SCALABILITY

Storage automatically scales with application requirements — from semester one to a decade of archives.



DURABILITY

S3 is designed for 99.99999999% (11 nines) durability of stored objects across multiple facilities.



SECURITY

IAM, bucket policies, encryption and access controls protect sensitive student data.



AVAILABILITY

Files can be accessed whenever authorized users need them, from anywhere.

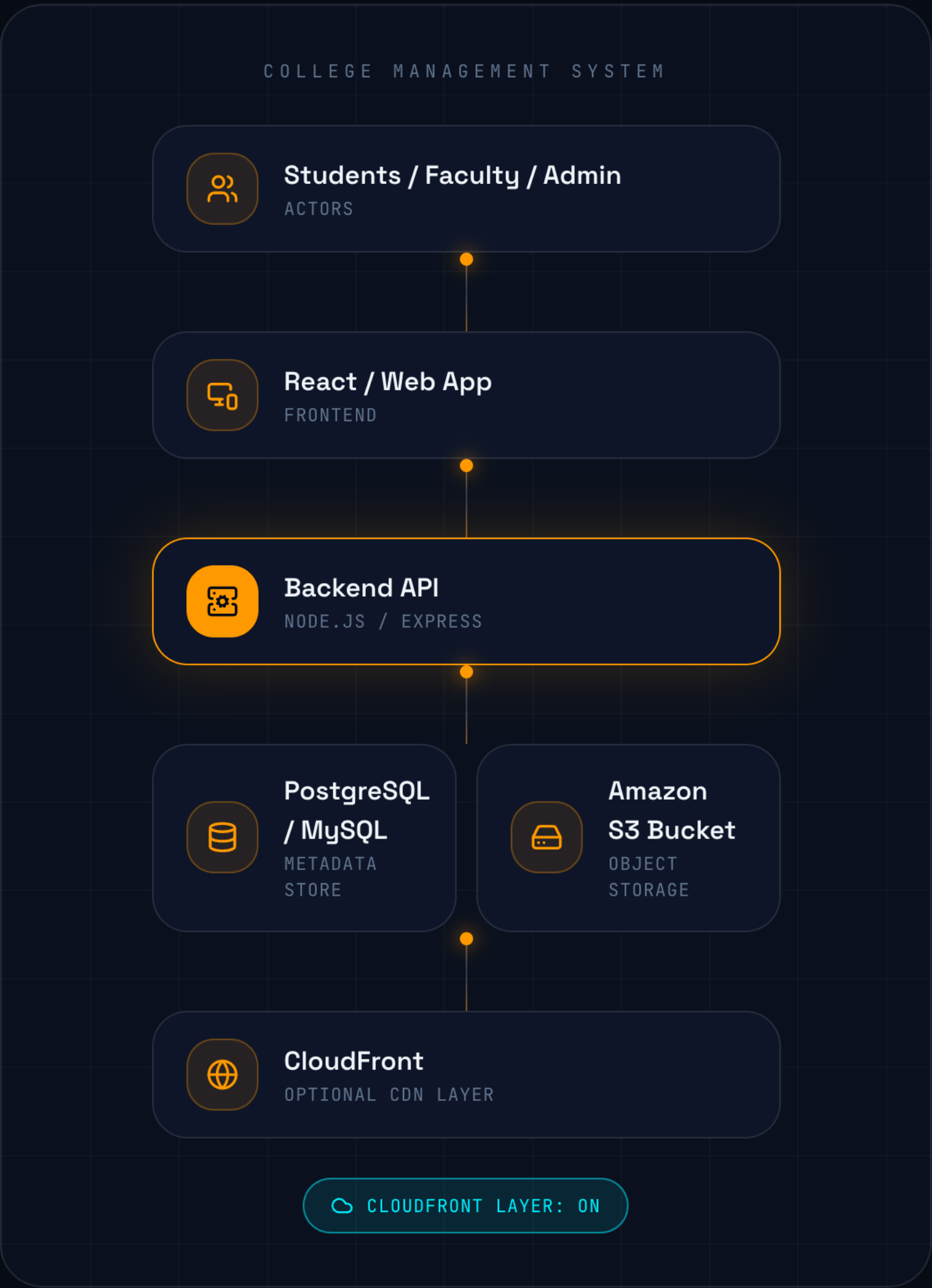


COST EFFICIENCY

Pay for storage and requests according to usage — no physical infrastructure to buy or maintain.

End-to-End System Architecture

Click any component to inspect its role. Watch data pulse through the pipeline.



Backend API

Handles authentication, validation, business logic and all communication with AWS. Issues presigned URLs.

Authenticates every request

Generates presigned URLs

Writes metadata to DB

TECHNICAL DEEP DIVE

Basic

Technical

The website shows pages, the server checks who you are, the database remembers information about each file, and S3 stores the files themselves.

Buckets, Objects & Keys

S3 is object storage — not a traditional hierarchical file system.



A bucket holds objects. Each object has a unique object key and metadata. The folders below are part of the key name — not real directories.

college-management-prod		PRIVATE
students/2026/1001/profile.jpg	486 KB	
students/2026/1001/aadhaar.pdf	1.1 MB	
students/2026/1001/certificate.pdf	2.4 MB	
assignments/CS101/1001/assignment1.pdf	890 KB	
faculty/204/lecture-video.mp4	184 MB	
events/2026/fest/day1-044.jpg	3.2 MB	

OBJECT INSPECTOR

 **certificate.pdf**

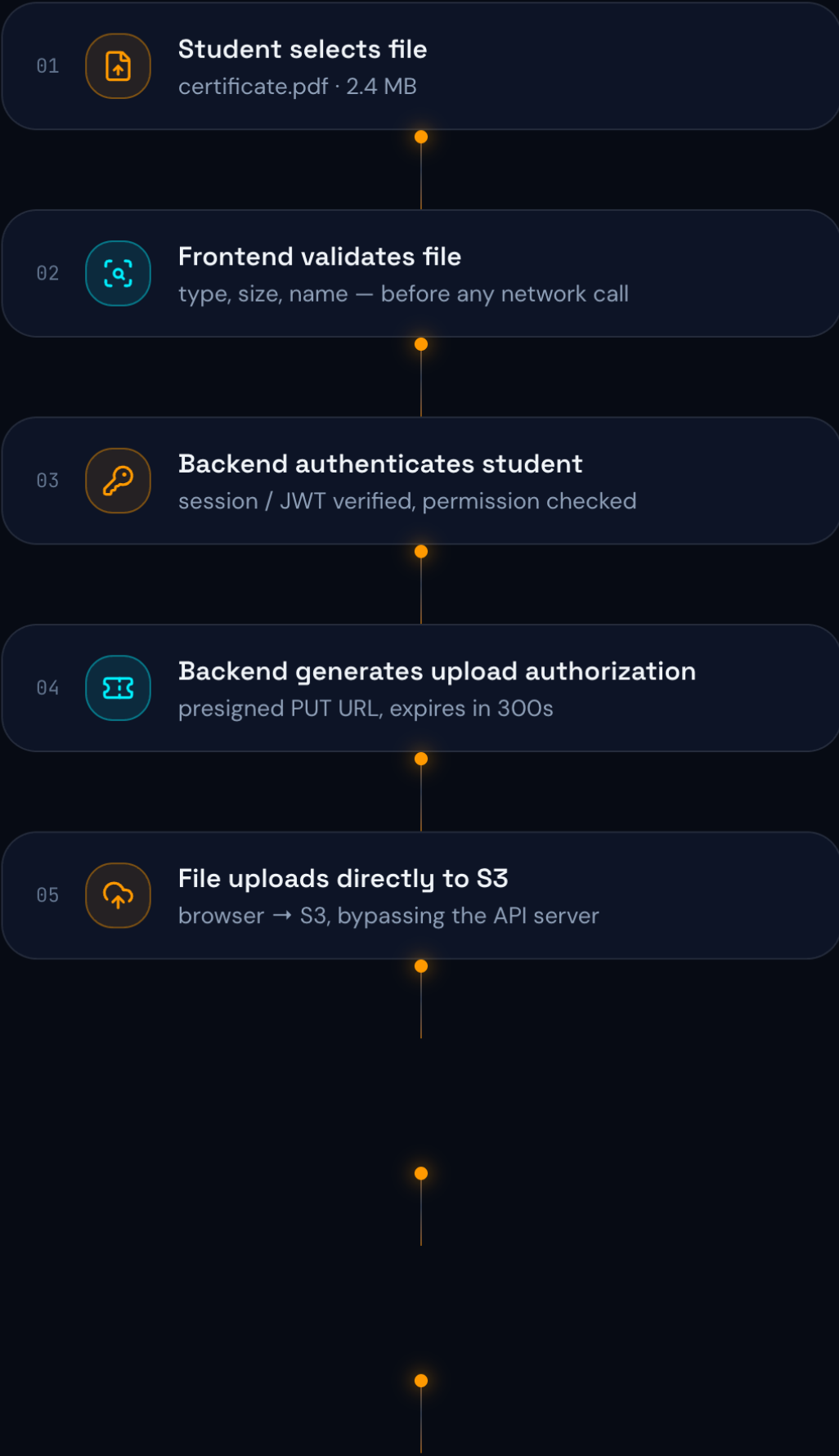
Key	students/2026/1001/certificate.pdf
Size	2.4 MB
Content-Type	application/pdf
Uploaded	10 Sept 2026
Storage Class	STANDARD

How Does a Student Upload a Certificate?

How Does a Student Upload a Certificate?

The browser uploads straight to S3 — AWS credentials never touch the frontend.

The key mechanism is the presigned URL: temporary upload authorization created by the backend, usable for a few minutes, for one specific object key only.



 TECHNICAL DEEP DIVE

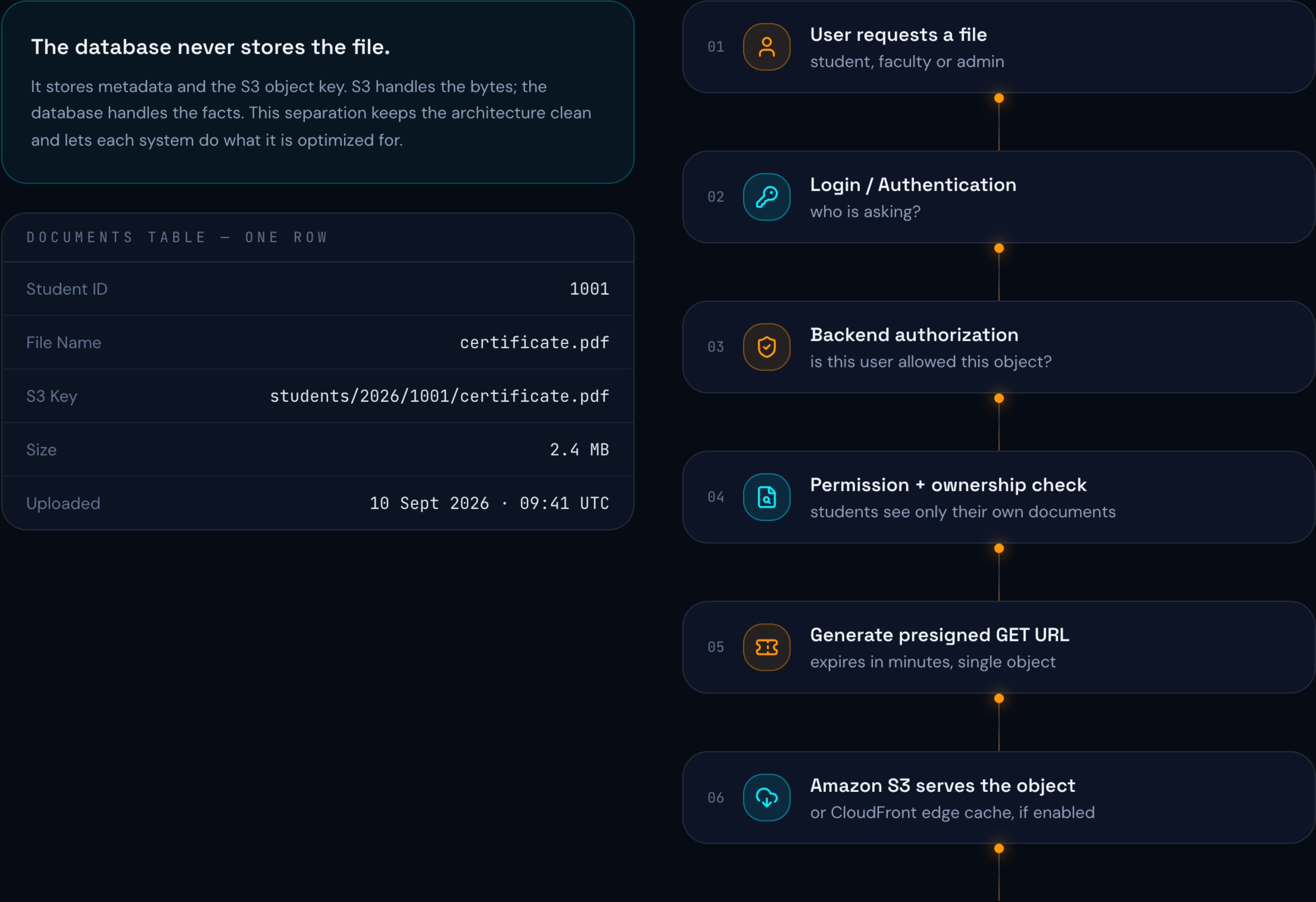
 Basic

 Technical

Instead of sending the file through the college server, the server gives the browser a special one-time pass, and the browser delivers the file straight to S3.

How Does a Student or Faculty Member Access a File?

Every download is authenticated, authorized, temporary — and auditable.



Security: The Most Important Layer

Student identity documents are sensitive — six layers stand between them and the internet.

L1

 **Authentication**
Student login · Faculty login · Admin login

L2

 **Authorization**
Role-based permissions · Students → own files only · Faculty → permitted academic files

L3

 **IAM Roles & Policies**
IAM roles, not root keys · Principle of least privilege

L4

 **Bucket Security**
Block all public access · Carefully scoped bucket policies · Sensitive files stay private

L5

 **Encryption**
SSE-S3 / SSE-KMS at rest · HTTPS / TLS in transit

L6

 **Temporary Access**
Presigned URLs · Short expiration times



 BAD ARCH

Vulnerable

 **Public S3 Bucket**
No access control restrictions

↓

 **Anyone with the URL**
Anonymous internet scrapers

↓

 **Student Aadhaar Exposed**
Confidential identity documents leaked

↓

 One leaked link = Major Incident

 SECURE ARCH

Zero Trust

 **Authenticated User**
Verified student/faculty credentials

↓

 **Backend Authorization**
RBAC validates student permissions

↓

 **Temporary Presigned URL**
Cryptographic token expires in 15m

↓

 **Private S3 Object**
Encrypted at rest, never public

↓

Verified, temporary & fully logged

What Happens If a File Is Accidentally Deleted?

Versioning keeps history. Lifecycle policies quietly move old files to cheaper storage.

 **S3 Versioning**

certificate.pdf — every overwrite kept, nothing lost

 Version 3

CURRENT

 Version 2

RECOVERABLE

 Version 1

RECOVERABLE

With versioning enabled, an accidental overwrite or delete can be rolled back to a previous version.

 **Lifecycle Policies**

Automatic tier transitions as files age

 **S3 Standard**
Active semester documents

FREQUENT ACCESS

 **S3 Standard-IA**
Previous semesters, rarely opened

AFTER INACTIVITY

 **S3 Glacier**
Alumni records, compliance archives

LONG-TERM ARCHIVAL

→ **Cross-Region Replication:** S3 can automatically replicate mission-critical documents to a secondary AWS region for complete disaster recovery posture.



 TECHNICAL DEEP DIVE

Basic

Technical

S3 can keep old copies of files and automatically move rarely-used files to cheaper storage, like moving old records from a desk drawer to a store room.

Can It Handle Thousands of Students?

Object storage separates file storage from application servers — growth stops being a hardware problem.

📈 Scales Automatically Without New Hardware

100 students
1,000 students
10,000 students
Large academic archive

From hundreds to tens of thousands of concurrent students, Amazon S3 absorbs the burst traffic — the college never has to buy or rack another physical server.

🌐 CLOUDFRONT FOR HOT CONTENT



CloudFront caches popular files at edge locations closer to users.

📁 MULTIPART UPLOAD

Lecture.mp4 · 850 MB 

Part 1

Part 2

Part 3

Part 4

Large lecture videos split into parallel parts — if one part fails, only that chunk retries.

How Much Does It Cost?

There is no single fixed price — S3 is usage-based, and that is the point.

MONTHLY COST ≈

Storage

+

Requests

+

Data Transfer

+

Retrieval

+

Optional Services

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There is no single fixed price — S3 is usage-based, and that is the point.

MONTHLY COST ≈

Storage

+

Requests

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Data Transfer

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Retrieval

+

Optional Services

 Amount of stored data

 PUT request count

 GET request count

 Data transfer out

 Storage class mix

 Retrieval frequency

 AWS region

 CloudFront usage

 The honest answer: estimate the college's real usage — gigabytes stored, uploads and downloads per month, retention policy — then compute the bill from **current AWS regional pricing**. Always verify pricing before deployment; it varies by region and changes over time.

College Cloud Storage — Live Simulation

A working feel of the S3 console: browse prefixes, inspect objects, upload a file.

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👤 FRONTEND SIMULATION — NO AWS CONNECTION, NO CREDENTIALS

TOTAL STORAGE
68 GB

OBJECTS
6,631

STUDENTS
1,726

DOCUMENTS
4,472

IMAGES
1,540

VIDEOS
619

🗃️ All objects

🎓 Students3,250

📖 Assignments4,102

🏷️ Certificates2,318

👤 Faculty1,406

📷 Events1,410

🏠 college-management-prod

🗃️ ARCHITECTURE VIEWUpload File

📄 students/2026/1001/certificate.pdf2.4 MB

📄 assignments/CS101/1001/assignment1.pdf890 KB

📄 faculty/204/lecture-video.mp4184 MB

📄 students/2026/1001/profile.jpg486 KB

📄 events/2026/fest/day1-044.jpg3.2 MB

What Happens When I Click Upload?

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Watch certificate.pdf travel from a student's laptop into the bucket — step by step, with realistic latency.

▶ Trace the request

	Browser selects certificate.pdf	0 ms
	Backend authenticates student	~45 ms
	Backend checks file type & size	~5 ms
	Backend creates presigned upload URL	~120 ms
	Browser uploads directly to S3	~850 ms
	S3 stores the object	~200 ms
	Backend stores metadata	~35 ms
	Student receives success response	~20 ms



Advantages vs Limitations

A good engineer names the trade-offs, not just the wins.

ADVANTAGES

- ✓ Highly scalable
- ✓ Durable storage
- ✓ Secure access control
- ✓ Easy backup & versioning
- ✓ Suitable for large files
- ✓ Supports lifecycle policies
- ✓ Integrates with web applications
- ✓ Reduces physical infrastructure

LIMITATIONS

- ⚠ Requires internet connectivity
- ⚠ AWS costs must be monitored
- ⚠ Incorrect permissions can expose data
- ⚠ Architecture requires proper IAM configuration
- ⚠ Data transfer / retrieval costs can matter
- ⚠ Vendor dependency

Store files in S3.

Store metadata in the database.

**Control access through authentication,
authorization and IAM.**

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RESULT

A scalable, secure and maintainable cloud storage architecture
for college applications.

THANK YOU_